# Sharon Zhou

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## Education

**University of Chicago** *2023 – Present*
*Ph.D. in Evolutionary Biology* (Advisor: Dr. David Jablonski)

**University of Chicago** *2019 – 2023*
*B.A. in Mathematics with honors, magna cum laude*

## Publications

- **Zhou, S.**, S.M. Edie, K.S. Collins, N.M.A. Crouch, and D. Jablonski. 2023. Cambrian origin but no early burst in functional disparity for Class Bivalvia. *Biology Letters* 19(5): 20230157. doi:10.1098/rsbl.2023.0157.
- Cossaboom, C., and **S. Zhou**. 2023. Hecke nilpotency for modular forms mod 2 and an application to partition numbers. *The Ramanujan Journal* 62: 899–923. doi:10.1007/s11139-023-00720-6.
- **Zhou, S.**, and L.E. Hunter. 2022. Digest: Evolution of plasticity and its potential role in the decline of specialists. *Evolution* 76(12): 3067–3070. doi:10.1111/evo.14633.

## Published abstracts

- **Zhou, S.**, S.M. Edie, K.S. Collins, and D. Jablonski, 2025. Latitudinal trends in morphological complexity: gradients, filters, and discordance with species richness. Geological Society of America Annual Meeting, San Antonio, TX. *Geological Society of America Abstracts with Programs* 57(6): Abstract 58-8. doi:10.1130/abs/2025AM-7590.
- **Zhou, S.**, K.S. Collins, S.M. Edie, and D. Jablonski, 2024. Loss and rebound of morphological and functional disparity after a non-selective mass extinction at the K/Pg boundary (Family Veneridae, Bivalvia). 12$^{th}$ North American Paleontological Convention Abstracts. University of Michigan Papers on Paleontology 39: 461-462.
- **Zhou, S.**, S.M. Edie, K.S. Collins, and D. Jablonski, 2023. Early burst in form but not function in Early Paleozoic bivalves: Distinct trends in taxonomy, morphology and ecology. Geological Society of America Annual Meeting, Abstract 197-8.
- **Zhou, S.**, K.S. Collins, N.M.A. Crouch, S.M. Edie, and D. Jablonski, 2022. No early burst in ecological diversity in Class Bivalvia: Integrating taxic and phylogenetic approaches. Geological Society of America Abstracts with Programs, Annual Meeting, Abstract 13-10.
- **Zhou, S.**, K.S. Collins, S.M. Edie, and D. Jablonski, 2021. Disparification before, during, or after diversification? Testing an Early Burst hypothesis for functional diversity in Early Paleozoic bivalves. Geological Society of America Abstracts with Programs, Annual Meeting 53: Abstract 141-11.

## Additional conference presentations

- **Zhou, S.**, S.M. Edie, and D. Jablonski. Bivalve shells as space–time records: what ornament reveals about development. Poster, NITMB workshop *Representing and Learning Morphology and Shape Dynamics from Biological Data*, Chicago, IL, October 2026.
- **Zhou, S.**, L. Hayward, W. Mawass, and J. J. Berg. Linked neutral diversity under stabilizing selection: A two-locus moments approach. Poster, Midwest Population Genetics Meetings, Chicago, IL, August 2026.
- **Zhou, S.**, S.M. Edie, K.S. Collins, and D. Jablonski. Parallel decoupling of species richness and morphological complexity across latitude in marine bivalves. Oral presentation, Evolution 2026, Cleveland, OH, June 2026.
- **Zhou, S.**, L. Hayward, W. Mawass, and J. J. Berg. Beyond purifying selection: A quantitative genetic model of background selection due to stabilizing selection. Midwest Population Genetics Conference, Minneapolis, MN, August 2025.
- Cossaboom, C., and **S. Zhou**. Hecke nilpotency for modular forms mod 2 and an application to partition numbers. Joint Mathematics Meetings, Boston, MA, January 2023.

## Selected local presentations

- **Zhou, S.**, S.M. Edie, K.S. Collins, and D. Jablonski. Clade sorting decouples morphological complexity and taxonomic richness across latitude in marine bivalves. Oral presentation, Darwinian Cluster annual retreat, University of Chicago, September 25, 2026.
- **Zhou, S.**, and D. Jablonski. Towards a generative morphospace of bivalve ornamentation. Oral flash talk, informal evo-devo seminar during Chris Lowe's visit, University of Chicago, September 24, 2026.
- **Zhou, S.**, S.M. Edie, K.S. Collins, and D. Jablonski. Species richness and morphological complexity decouple across latitude in marine bivalves of two ocean basins. Oral presentations, University of Chicago: Geophysical Sciences Expo, May 22, 2026; Committee on Evolutionary Biology retreat, May 1, 2026.

## Research Experience

**Graduate student researcher** | *Jablonski Lab, University of Chicago* *2023 – Present*

- Develop a novel morphological complexity score for bivalve ornamentation.
- Investigate the spatial and temporal structure of biodiversity and macroevolutionary trends with quantitative modeling approaches.

**Graduate student researcher** | *Berg Lab, University of Chicago* *2024 – Present*

- Investigate how stabilizing selection shapes patterns of genomic diversity, using forward simulations and analytical models to advance background selection theory.

## Teaching Experience

**Teaching Assistant**, Quantitative Genetics for the 21$^{st}$ Century, *University of Chicago* Spring 2026

- Supported a mixed undergraduate/graduate class in bridging quantitative genetic theory - including the Breeder's equation, causal inference, and GWAS - with hands-on R implementation through coding assignments. Moderated in-class paper discussions on the intersections of race, ancestry, and complex traits.

**Teaching Assistant**, Biological Evolution, *University of Chicago* Winter 2026

- Facilitated active-learning discussion sections for non-majors and majors, guiding students through primary literature analysis and evolutionary theory. Co-developed inquiry-based curriculum for a Field Museum experiential learning module, helping students synthesize morphological data with evolutionary hypotheses.

## Awards and Grants

- SSE Graduate Student and Postdoc Registration Waiver, Society for the Study of Evolution (2026); registration support for Evolution 2026.
- GSA North-Central Section Student Travel Grant, Geological Society of America (2025)
- Quad Faculty Research Grant, University of Chicago (2021 – 2022) *Supervisor*: David Jablonski
- Dean's Fund for Undergraduate Research, University of Chicago (2022) *Supervisor*: David Jablonski
- Phi Beta Kappa Society (2022)
- Enrico Fermi Scholar Physical Sciences Collegiate Divisional Honors, University of Chicago (2022)
- Robert Maynard Hutchins Scholar, University of Chicago (2021)
- Dean's List, University of Chicago (2020 – 2023)

## Workshops

- Representing and Learning Morphology and Shape Dynamics from Biological Data. October 2026, NITMB, Chicago, IL.
- Contemporary Challenges in Large-Scale Sequence Alignments and Phylogenies. August 2025, IMSI, University of Chicago, Chicago, IL
- Meeting of the Society for Modeling and Theory in Population Biology. June 2025, NITMB, Chicago, IL
- 2024 AI+Science Summer School. July 2024, IMSI, University of Chicago, Chicago, IL

## Service

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| Darwinian Cluster student representative, University of Chicago | 2025 – Present |
| Sewall Wright Lecture organizing committee, University of Chicago | 2024 – Present |

## Skills

**Computational:** Programming experience in R and Python; LaTeX; Adobe Photoshop, Excel